

LECTURE 03-THE BIOLOGY OF MIND

1. What is the Mind?

Do you have a mind? What is it? Where is it?

- The mind is not an object you can touch.
- The mind is what your brain does.
- It is created through the brain's information processing.

Functions of the Mind

Your mind enables:

- Thinking
- Decision-making
- Memory
- Emotions
- Imagination
- Perception (what you see, hear, feel, taste, smell)

In simple words:

📁 Your mind is your inner world, created by the brain's activity.

2. The Brain – The Central Command Center

The brain is the control system that regulates nearly everything in the body.

Major functions of the brain

- Thoughts (reasoning, planning, problem-solving)
- Memory (storing and recalling information)

- Learning (forming new connections between neurons)
- Emotions (fear, happiness, anger, love)
- Movement (coordination, balance)
- Senses (interpreting visual, auditory, taste, touch, smell signals)
- Automatic functions
- Heartbeat
- Breathing
- Digestion
- Sleep cycles
- Stress response
- Hormone regulation (e.g., puberty)

Mini-diagram (described):

Imagine a control center filled with switches — each switch manages one body function. That control center is your brain.

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3. The Nervous System

The nervous system is the body's communication highway.

Two main parts:

1. Central Nervous System (CNS)
 - Brain
 - Spinal cord
2. Peripheral Nervous System (PNS)
 - All nerves outside the brain and spinal cord
 - Connects the CNS to the body (muscles, organs, skin)

Purpose:

To send messages rapidly throughout the body using electrical and chemical signals.

4. Neurons – The Brain’s Building Blocks

- A neuron is a nerve cell that sends and receives information.
- You have ~86 billion neurons in your brain.

Parts of a neuron

- Dendrites – receive messages
- Cell body (soma) – processes information
- Axon – sends messages
- Synapse – tiny gap where chemical signals pass to the next neuron

Example

When you touch a hot cup:

- Sensory neurons send a pain signal → brain
- Brain processes it
- Motor neurons send message → move your hand away

5. The Brain’s Organization (3 Major Parts)

A. The Brainstem

The oldest and most basic part of the brain.

Controls survival functions.

Parts of the Brainstem

1. Medulla – controls heartbeat and breathing
2. Pons – controls sleep, facial expressions, and movement coordination
3. Reticular formation – controls alertness and arousal
4. Thalamus – the “sensory relay station”

- Sends sensory information (except smell) to higher brain regions

Mini-diagram (described):

A stalk-like structure at the bottom of the brain — that's the brainstem.

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B. The Limbic System

The emotional center of the brain.

Key Structures

1. Amygdala – emotion (especially fear and aggression)
 - Example: Your amygdala activates when you see a snake.
2. Hippocampus – memory formation
 - Example: Helps you remember your first day of school.
3. Hypothalamus – regulates hunger, thirst, temperature, hormones
 - Commands the pituitary gland (master hormone gland)
4. Pituitary gland – releases hormones controlling growth, stress, reproduction

Mini-diagram (described):

Small structures deep inside the brain forming a ring — that is the limbic system.

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C. The Cerebral Cortex

The newest, most advanced part of the brain.

Responsible for complex thinking, planning, language, and consciousness.

Functions

- Logic
- Creativity

- Language
 - Moral judgement
 - Planning
 - Problem-solving
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6. Case Study: Phineas Gage

The most famous case in neuroscience.

The Accident

- In 1848, a metal rod shot through Gage's cheek and out the top of his skull.
- It destroyed much of his frontal lobes.

What happened after?

- He survived and could speak, walk, and think.
- BUT his personality changed completely:
- Became irritable
- Impulsive
- Dishonest
- Lost ability to manage emotions
- Could not plan properly

His friends said he was "no longer Gage."

Why is this important?

It showed that:

- The frontal lobe controls personality, decision-making, and emotion regulation.
 - Damage to the brain can change who we are, not just how we think.
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7. Language and the Brain (Aphasia)

Aphasia = language impairment from brain damage.

There are 3 important language areas:

1. Broca's Area (Left Frontal Lobe)

Function

- Controls speaking
- Helps form words
- Coordinates speech muscles

Damage → Broca's Aphasia

- Person understands speech
 - BUT cannot speak fluently
 - Speech becomes broken or slow
 - Example: "Walk... dog... store..."
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2. Wernicke's Area (Left Temporal Lobe)

Function

- Understands language
- Gives meaning to words

Damage → Wernicke's Aphasia

- Person can speak fluently
- BUT speech is meaningless

- Example sentence from a patient:

“Mother is away her working her work to get her better...”

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3. Angular Gyrus

Function

- Converts written text → auditory form
- Required for reading aloud

Damage

- Person can speak
- Person can understand speech
- BUT cannot read

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8. How Reading Words Works (Flow of Processing)

1 → Visual Cortex

Sees written words.

2 → Angular Gyrus

Converts written words into sound-like code.

3 → Wernicke's Area

Interprets meaning.

4 → Broca's Area

Plans speech.

5 → Motor Cortex

Moves the mouth to talk.

Example:

When reading the word “cat” aloud, your brain follows this exact pathway.

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9. Thinking Styles (Left vs Right Brain Tendencies)

Not scientifically rigid, but helpful to understand preferences.

Intuitive / Creative Style

- Thinks in images
- Likes stories, patterns
- Prefers hands-on learning
- Enjoys creative hobbies (art, music, writing)

Logical / Analytical Style

- Breaks problems into steps
- Prefers facts and sequences
- Likes instructions and guides
- Enjoys puzzles, math, strategy games

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10. How to Enhance Brain Fitness (Based on Your Slides)

A. Sound Sleep

- Aim for 7.5 hours per night
- Sleep boosts memory, learning, mood, and focus
- Example: After studying, sleep helps store information more deeply.

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B. Eat Well

Foods that support brain health:

- Nuts
- Blueberries
- Whole grains
- Avocados
- Fish oil
- Broccoli
- Tomatoes
- Seeds
- Water

Nutrients important for the brain:

- Antioxidants (protect brain cells)
 - Amino acids (help with neurotransmitters)
 - Vitamin E (brain protection)
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C. Write Things Down

- Helps clear your mind
 - Reduces stress
 - Improves memory
 - Great habit: Write tomorrow's plan the night before
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D. Slow Down

- Focus on one task at a time
- Spend time with people
- Build friendships

- Be mindful
 - Practise empathy
 - Helps your brain stay calm and organized
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E. Breathe

- Deep breathing reduces stress
 - Improves oxygen flow to the brain
 - Helps focus and emotional balance
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Complete Summary (Quick Revision Section)

- Mind = what the brain does
- Brain controls everything: thoughts → emotions → senses → memory → body functions
- Neurons send electrical + chemical signals
- Brainstem = survival
- Limbic system = emotion, memory
- Cortex = thinking, planning, language
- Phineas Gage shows frontal lobe controls personality
- Broca's area = speaking
- Wernicke's area = understanding
- Angular gyrus = reading
- Brain fitness → sleep, nutrition, writing, slowing down, breathing